Test Case / Original Request

Run PTC-001 using k6. Execute the real performance test, monitor it, assess the security surface for the same job and configured target, analyze the results, and create the final PDF with links to the native report and raw results.

REAL Execution Outcome

Execution status: COMPLETED
Runner status: completed
Outcome details: REAL k6 execution completed. Threshold breaches, if any, are retained as performance findings.
Thresholds passed: True

Executive Summary

Application: Unknown
Scenario: Content Record Retrieval API — Load Check
Job ID: JOB-3efd8c42b9404ed290bcb1e2d2b95fd4
Overall Result: WARNING

The API responded consistently under load, but throughput dipped below the 1.0 req/s threshold, triggering a warning for performance. Infrastructure health was strong, with low CPU and memory usage and no alerts or anomalies. However, a security issue was flagged: the absence of the Strict-Transport-Security (HSTS) header in the observed response, resulting in a security warning. No relevant historical run or SLA context was available; findings are based only on current run and static performance/security thresholds.

Execution Profile and Artifacts

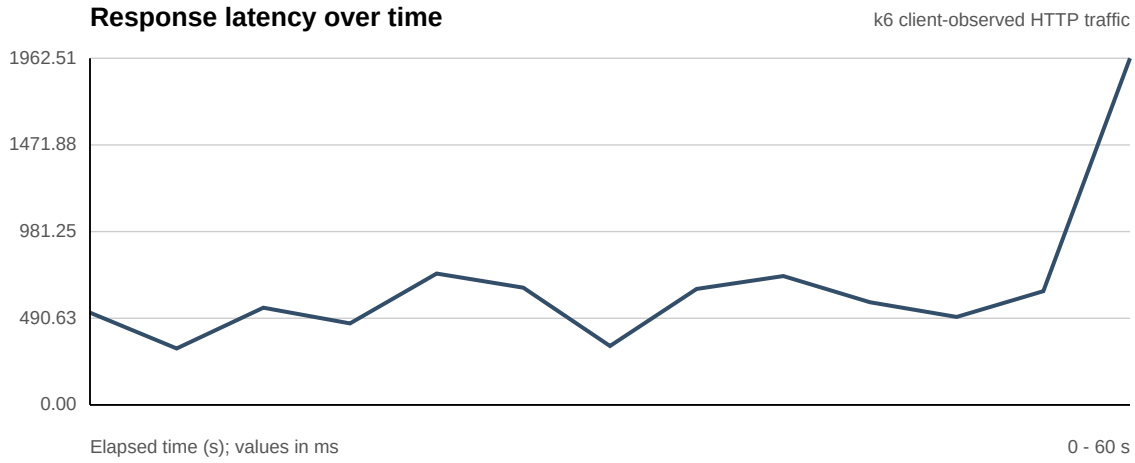
Execution mode: REAL
Workload profile:
- tool: k6
- target_url: https://jsonplaceholder.typicode.com
- endpoint: /posts/1
- method: GET
- users: 1
- max_rps: 1
- duration_seconds: 60
- ramp_up_seconds: 5
- expected_status_code: 200
- p95_threshold_ms: 5000
Artifact directory: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4
Artifact references:
- aligned_time_series: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_aligned_time_series.csv
- generated_script: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check.js
- k6_summary: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_summary.json
- measurement_metadata: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_measurement_metadata.json

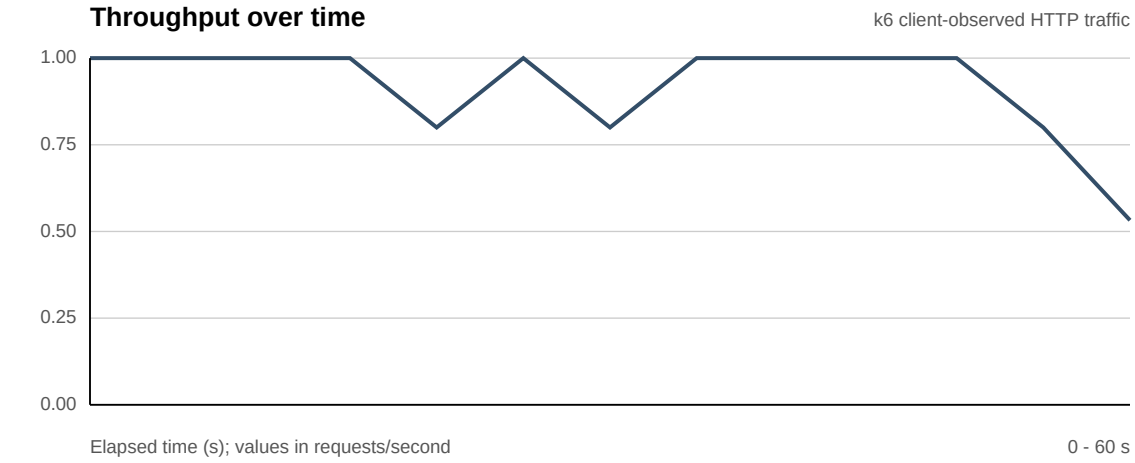
- monitoring_samples: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_runtime_samples.csv
- Native k6 HTML report
- Native report + raw results ZIP
- normalized_metrics: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_normalized_metrics.json
- Raw k6 results (JSON)
- runner_log: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_runner.log
- time_series: src/generated_scripts/k6 native reports/JOB-3efd8c42b9404ed290bcb1e2d2b95fd4/k6_PTC_001_Content_Record_Retrieval_API_Load_Check_time_series.csv

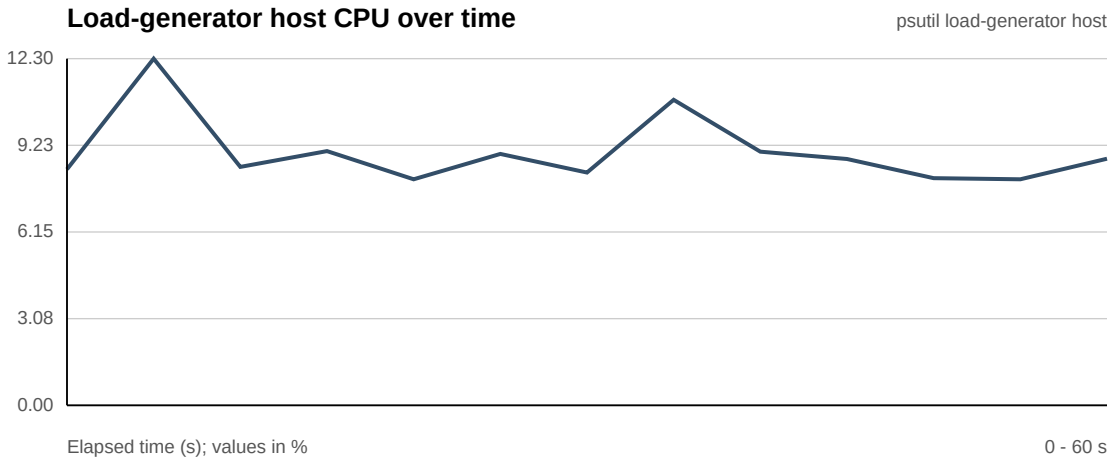
The interactive native HTML dashboard is linked as a separate artifact and is not embedded in this PDF.

Time-Series Charts (REAL Measurements)

Charts show measured values in five-second aligned buckets. Host CPU is from the load-generator machine, not the target.







Agent Activity Summary

Agent	*Summary*
Test Execution	Load test executed on /posts/1 (Unknown - Content Record Retrieval API — Load Check). Results: 58 total requests, 58 successful, 0 failed. Avg response time: 582.1222931034482 ms; 95th percentile: 1289.2938 ms; Max: 1962.508 ms. Throughput: 0.9380270893490049 TPS; Error rate: 0.0%.
Monitoring	CPU usage: 9.0% average, 18.0% peak; memory: 63.81% average, 64.7% peak; DB connections: not collected, TPS: 0.9380270893490049, Response time: 582.1222931034482 ms. Data source and scope: REAL: psutil load-generator host samples and k6 client-observed metrics. Measurement provenance: cpu_usage=psutil (load_generator_host); cpu_peak_usage=psutil (load_generator_host); memory_usage=psutil (load_generator_host); memory_peak_usage=psutil (load_generator_host); response_time=k6 (load_generator_observed_http_traffic); tps=k6 (load_generator_observed_http_traffic); db_connections=not_collected (No target-specific database telemetry provider is configured.). No alerts triggered.

Agent	*Summary*
Security Validation	Status: WARNING. Security score: 0%. Live security check: completed. Issue: Strict-Transport-Security was not present in the observed response; this single response does not establish the full origin policy. Issue: Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness. Issue: Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness. Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0} Control assessment evidence: - external_surface: assessed; basis=live_observation; evidence=['live-observation:JOB-3efd8c42b9404ed290bcb1e2d2b95fd4'] Evidence references: live-observation:JOB-3efd8c42b9404ed290bcb1e2d2b95fd4 Recommendation: Review whether an HSTS policy is appropriate for this HTTPS API origin.
Analysis	The API responded consistently under load, but throughput dipped below the 1.0 req/s threshold, triggering a warning for performance. Infrastructure health was strong, with low CPU and memory usage and no alerts or anomalies. However, a security issue was flagged: the absence of the Strict-Transport-Security (HSTS) header in the observed response, resulting in a security warning. No relevant historical run or SLA context was available; findings are based only on current run and static performance/security thresholds.
Reporting	A PDF report was generated, including all findings, recommendations, and an executive summary.

Per-API Findings (Test Execution)

- /posts/1 [WARNING / medium]
avg=582.1222931034482ms p95=1289.2938ms max=1962.508ms throughput=0.9380270893490049 req/s error_rate=0.0% http_status=200
* Throughput 0.9380270893490049 req/s is below the 1.0 req/s floor.
history: No historical context or RAG documents were available; assessment is based on the current static throughput threshold breach only.
recommendation: Review the minimum throughput requirements for this API—current throughput fell below the configured 1.0 req/s floor. No known incidents, SLA, or runbook context available for this scenario.

Infrastructure Health (Monitoring)

Status: PASS (monitoring_agent reported: OK)
CPU usage: 9.0% Memory usage: 63.81% DB connections: None TPS: 0.9380270893490049
Response time (monitoring): 582.1222931034482ms

Security Findings (Security Agent)

Status: WARNING Security score: 0%
Security result job ID: JOB-3efd8c42b9404ed290bcb1e2d2b95fd4.
Live security check: completed.
Issues:
- Strict-Transport-Security was not present in the observed response; this single response does not establish the full origin policy.
- Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness.
- Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness.
Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0}
Requested controls: external_surface
Assessed controls: external_surface
Control evidence and provenance:
- external_surface: assessed; basis=live_observation; evidence fields=['target_origin', 'endpoint', 'method', 'response_status_code', 'observed_security_headers']; references=['live-observation:JOB-3efd8c42b9404ed290bcb1e2d2b95fd4']
Evidence references: live-observation:JOB-3efd8c42b9404ed290bcb1e2d2b95fd4
Live observation: HEAD /posts/1; HTTP 200; elapsed 1018.3326250407845 ms; requests sent 1.
Observed response headers (observations only):
- cache-control: max-age=43200
- x-content-type-options: nosniff
Recommendations:
- Review whether an HSTS policy is appropriate for this HTTPS API origin.

Recommended Next Steps

- /posts/1: Review the minimum throughput requirements for this API—current throughput fell below the configured 1.0 req/s floor. No known incidents, SLA, or runbook context available for this scenario.
- Security: address the issues/recommendations listed above before next deployment.